Data Analysis
AMS315

Chapter 8

Confidence Interval Estimation

Learning Objectives

In this chapter, you learn:

- To construct and interpret confidence interval estimates for the mean and the proportion
- How to determine the sample size necessary to develop a confidence interval for the mean or proportion

Reminder: Make sure that you can solve all five examples in this chapter.

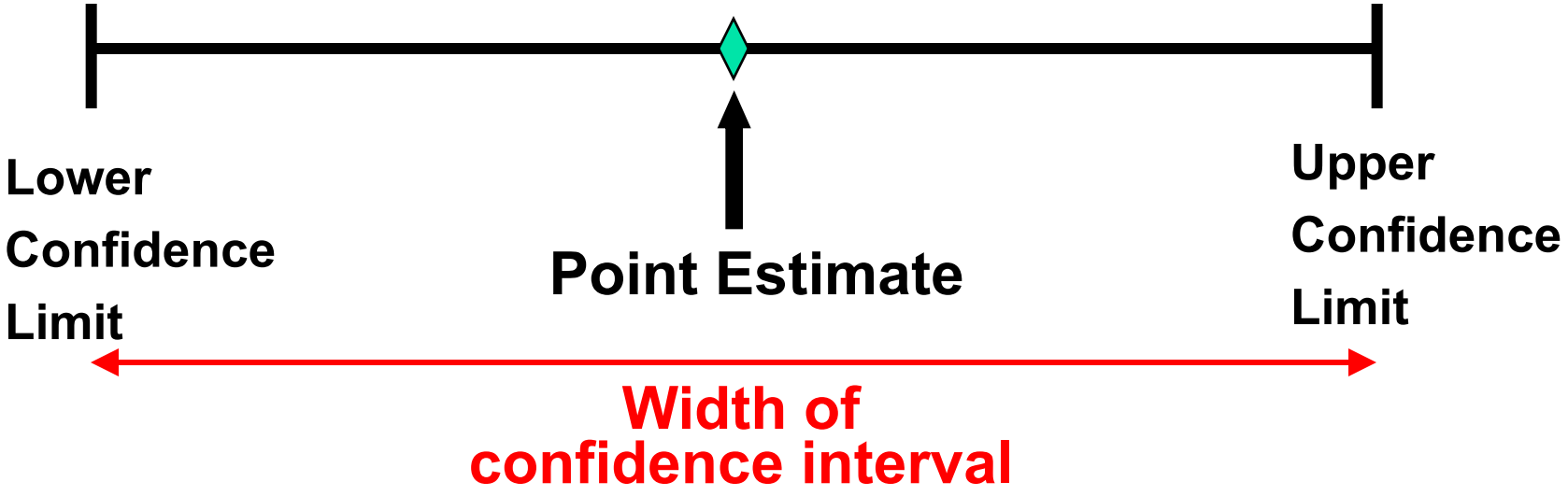
Confidence Intervals

Content of this chapter

- Confidence Intervals for the population mean μ
  - when Population Standard Deviation σ is Known
  - when Population Standard Deviation σ is Unknown
- Confidence Intervals for the population proportion π
- Determining the required sample size

Point and Interval Estimates

- A point estimate is a single number,
- a confidence interval provides additional information about variability



Point Estimates

We can estimate a Population Parameter …		with a Sample Statistic (a Point Estimate)
Mean	μ	$\bar{X}$
Proportion	π	p

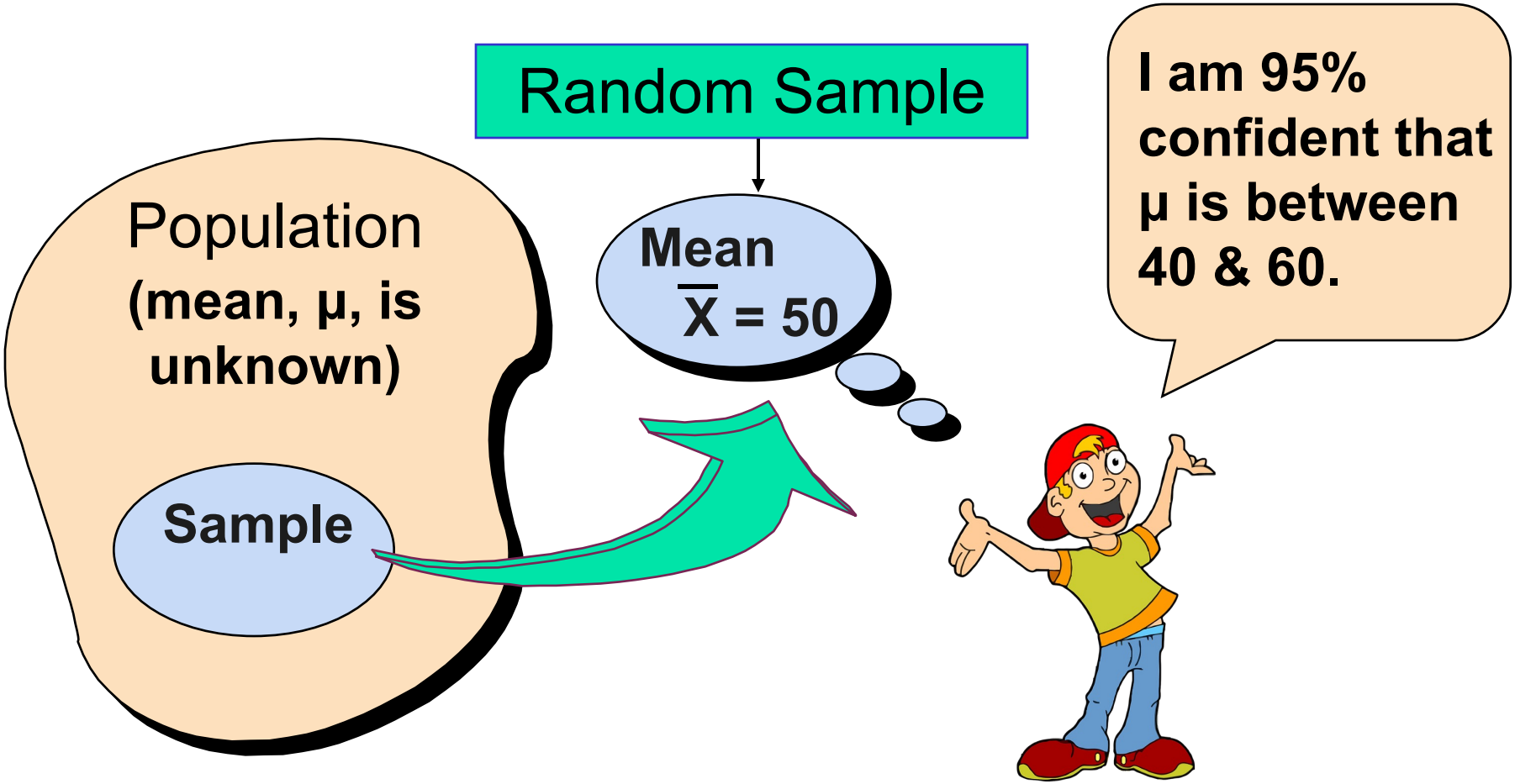
Confidence Intervals

- How much uncertainty is associated with a point estimate of a population parameter?
- An interval estimate provides more information about a population characteristic than does a point estimate
- Such interval estimates are called confidence intervals

Confidence Interval Estimate

- An interval gives a range of values:
  - Takes into consideration variation in sample statistics from sample to sample
  - Based on observations from 1 sample
  - Gives information about closeness to unknown population parameters
  - Stated in terms of level of confidence
    - Can never be 100% confident

Estimation Process



General Formula

- The general formula for all confidence intervals is:

Point Estimate ± (Critical Value)(Standard Error)

Confidence Level

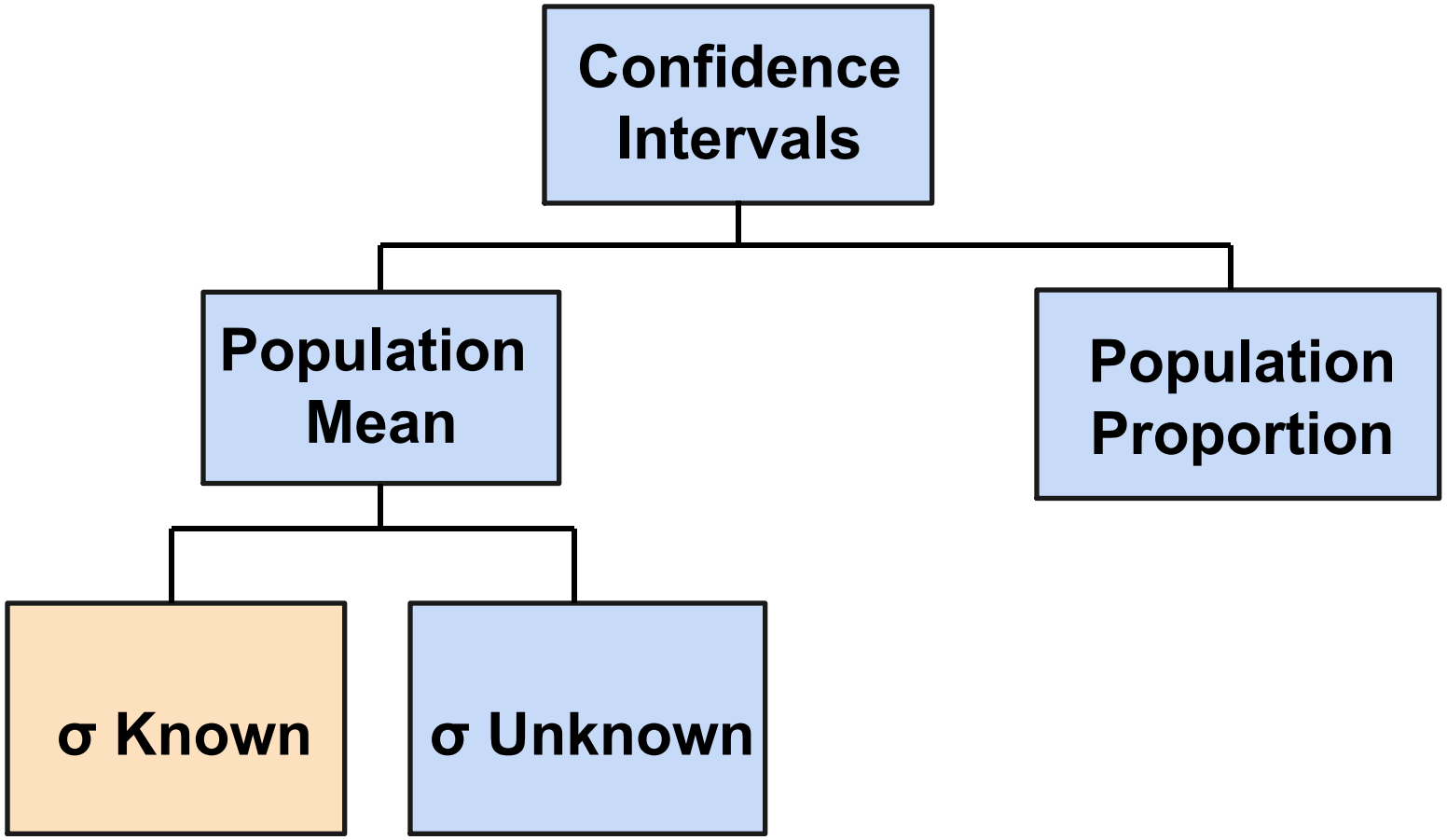
- Confidence Level
  - Confidence for which the interval will contain the unknown population parameter
- A percentage (less than 100%)

Confidence Level, (1-α)

(continued)

- Suppose confidence level = 95%
- Also written (1 - α) = 0.95
- A relative frequency interpretation:
  - In the long run, 95% of all the confidence intervals that can be constructed will contain the unknown true parameter
- A specific interval either will contain or will not contain the true parameter
  - No probability involved in a specific interval

Confidence Intervals



Confidence Interval for μ (σ Known)

- Assumptions
  - Population standard deviation σ is known
  - Population is normally distributed
  - If population is not normal, use large sample
- Confidence interval estimate:

$$\bar{X} \pm Z \frac{\sigma}{\sqrt{n}}$$

where $\bar{X}$ is the point estimate

Z is the normal distribution critical value for a probability of $\alpha/2$ in each tail

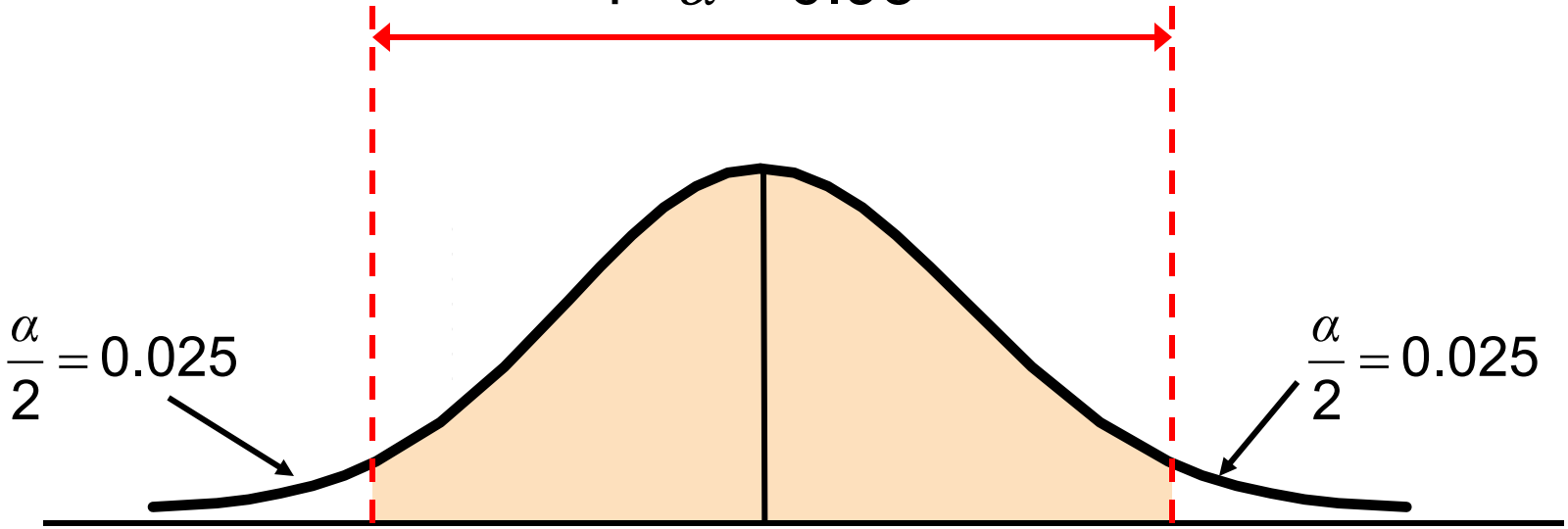
$\sigma/\sqrt{n}$ is the standard error

Finding the Critical Value, Z

$$Z = \pm 1.96$$

- Consider a 95% confidence interval:

$$1 - \alpha = 0.95$$



Z units: **Z= -1.96** **0** **Z= 1.96**

X units: **Lower Confidence Limit** **Point Estimate** **Upper Confidence Limit**

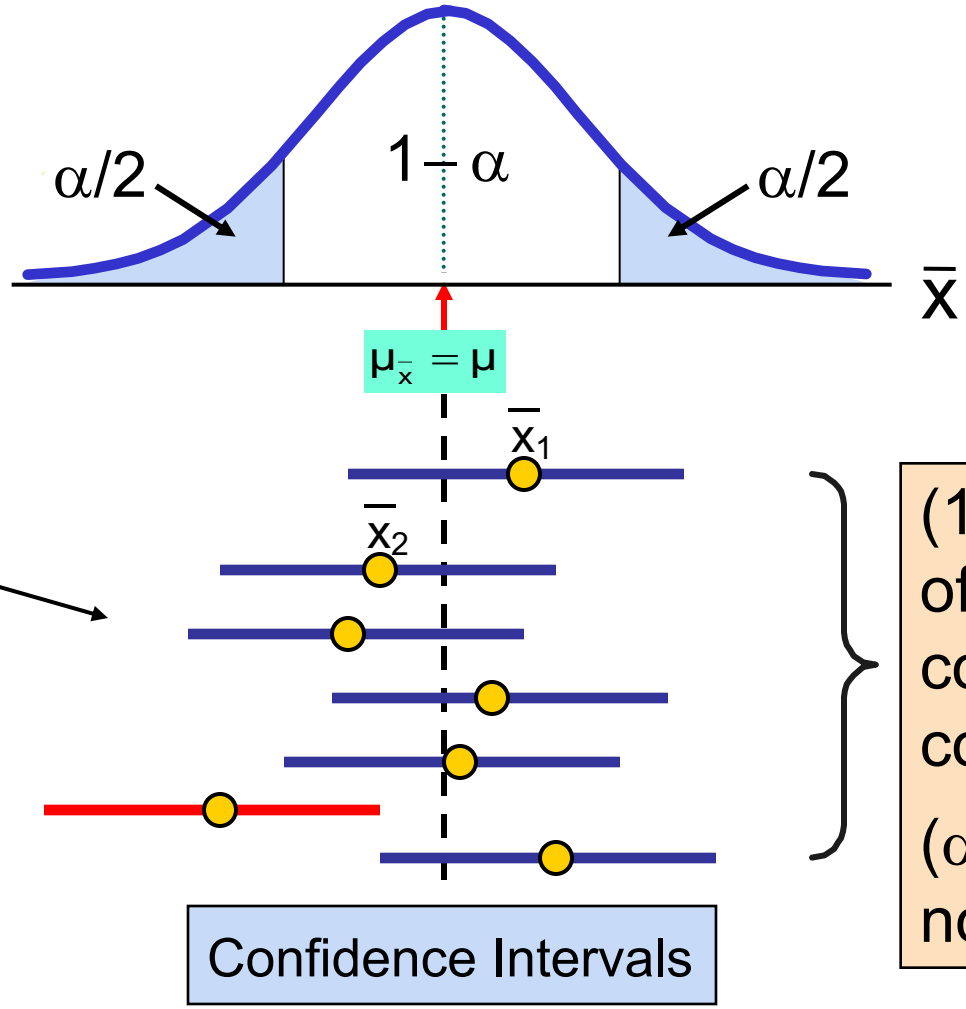
Common Levels of Confidence

- Commonly used confidence levels are 90%, 95%, and 99%

Confidence Level	*Confidence Coefficient,* $1-\alpha$	*Z value*
80%	**0.80**	**1.28**
90%	**0.90**	**1.645**
95%	**0.95**	**1.96**
98%	**0.98**	**2.33**
99%	**0.99**	**2.58**
99.8%	**0.998**	**3.08**
99.9%	**0.999**	**3.27**

Intervals and Level of Confidence

Sampling Distribution of the Mean



Intervals extend from

$$\bar{X} + Z\frac{\sigma}{\sqrt{n}}$$

to

$$\bar{X} - Z\frac{\sigma}{\sqrt{n}}$$

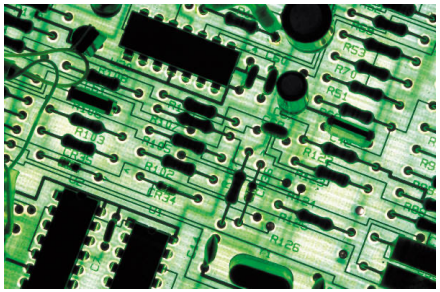
(1-α)x100% of intervals constructed contain μ;

(α)x100% do not.

Example

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms.

- Determine a 95% confidence interval for the true mean resistance of the population.

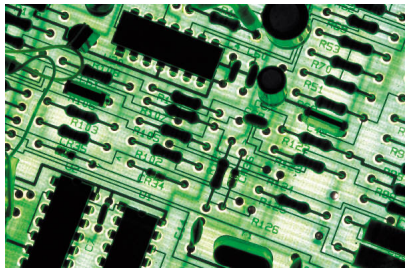


Example

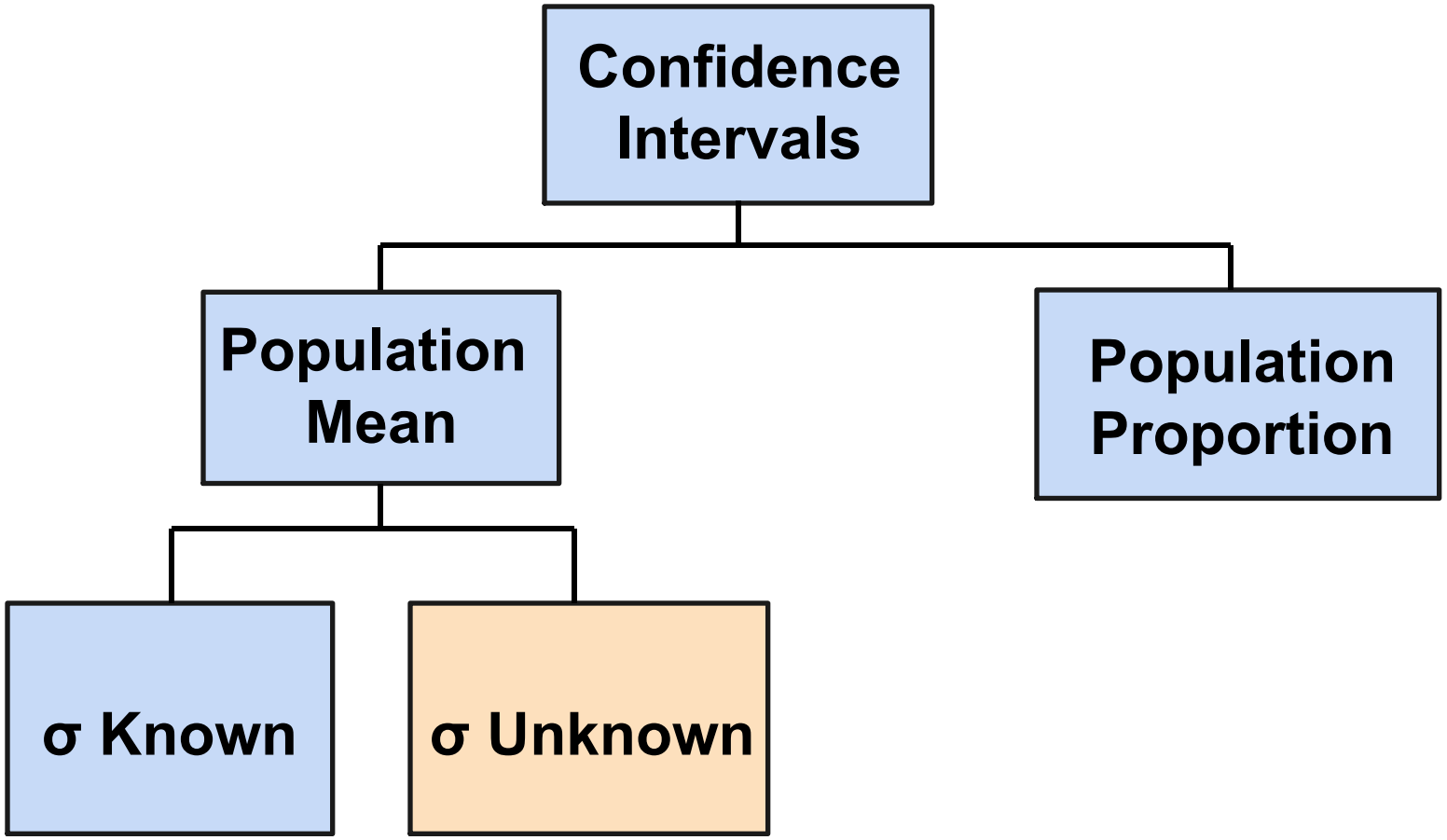
Solution:

Interpretation

- We are 95% confident that the true mean resistance is between 1.9932 and 2.4068 ohms
- Although the true mean may or may not be in this interval, 95% of intervals formed in this manner will contain the true mean



Confidence Intervals



Confidence Interval for μ (σ Unknown)

- If the population standard deviation σ is unknown, we can substitute the sample standard deviation, S
- This introduces extra uncertainty, since S is variable from sample to sample
- So we use the *t* distribution instead of the normal distribution

Confidence Interval for μ (σ Unknown)

(continued)

- Assumptions
  - Population standard deviation is unknown
  - Population is normally distributed
  - If population is not normal, use large sample
- Use Student's *t* Distribution
- Confidence Interval Estimate:

$$\bar{X} \pm t_{n-1} \frac{S}{\sqrt{n}}$$

(where t is the critical value of the t distribution with n -1 degrees of freedom and an area of α/2 in each tail)

Student's t Distribution

- The *t* is a family of distributions
- The *t* value depends on degrees of freedom (d.f.)
  - Number of observations that are free to vary after sample mean has been calculated

$$\text{d.f.} = n - 1$$

Degrees of Freedom (df)

Idea: Number of observations that are free to vary after sample mean has been calculated

Example: Suppose the mean of 3 numbers is 8.0

Let $X_1 = 7$
Let $X_2 = 8$
What is $\mathbf{X_3}$?

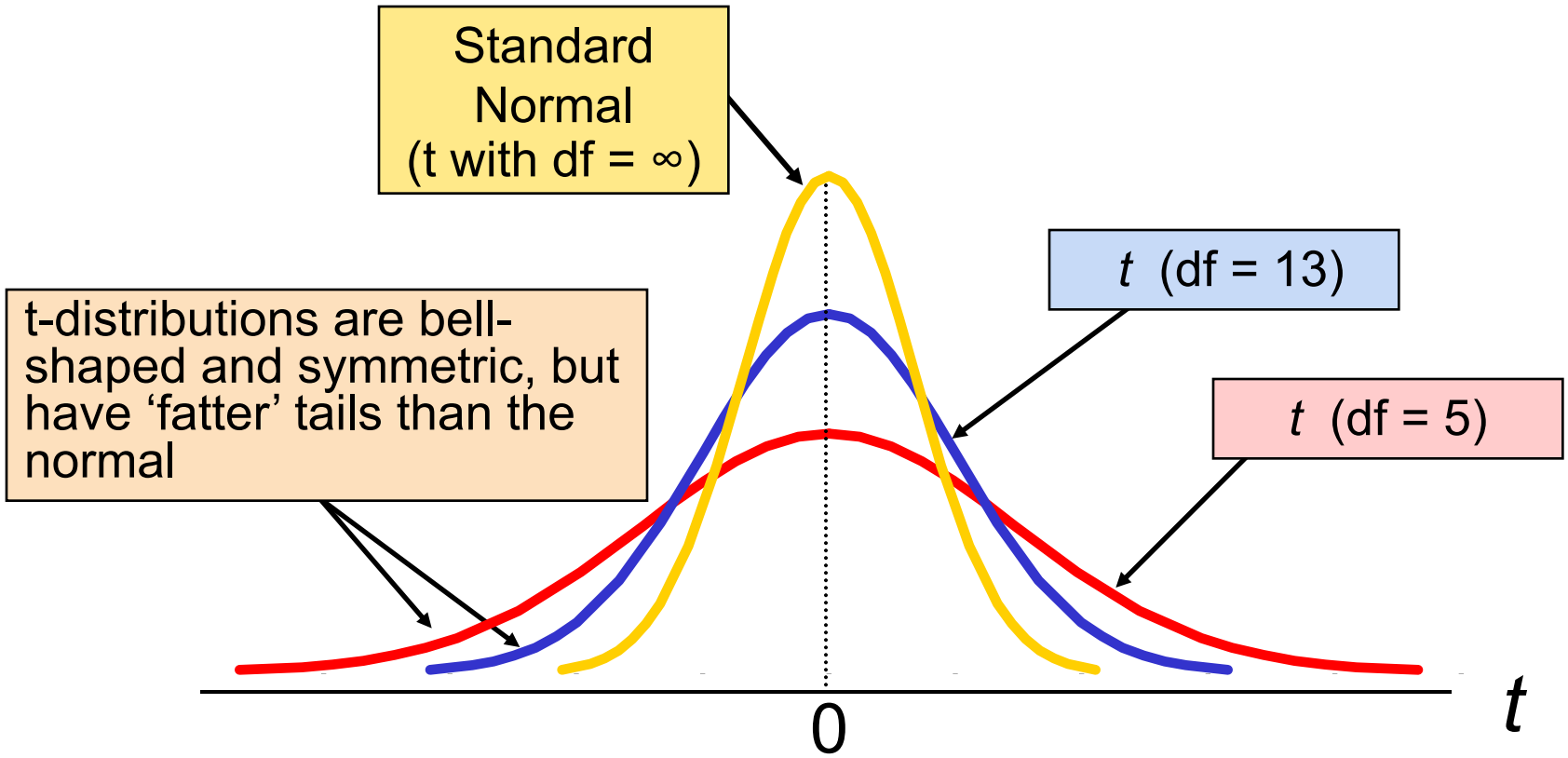
If the mean of these three values is 8.0,
then X_3 must be 9
(i.e., X_3 is not free to vary)

Here, n = 3, so degrees of freedom = n – 1 = 3 – 1 = 2

(2 values can be any numbers, but the third is not free to vary for a given mean)

Student's t Distribution

Note: $t \longrightarrow Z$ as n increases

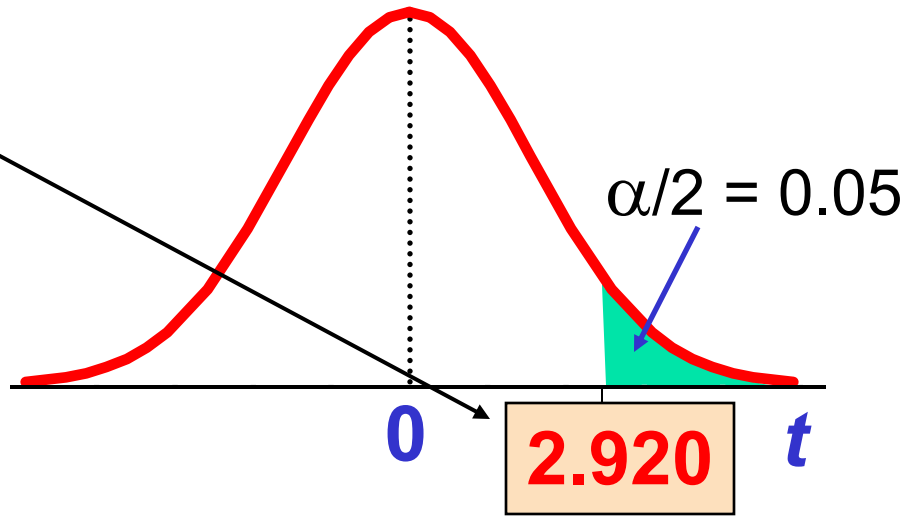


Student's *t* Table

	Upper Tail Area		
df	.25	.10	**.05**
1	1.000	3.078	6.314
2	0.817	1.886	**2.920**
3	0.765	1.638	2.353

The body of the table contains t values, not probabilities

Let: n = 3
df = n - 1 = 2
α = 0.10
$\alpha/2$ = 0.05



t distribution values

With comparison to the Z value

Confidence Level	*t* (10 d.f.)	*t* (20 d.f.)	*t* (30 d.f.)	Z
0.80	1.372	1.325	1.310	1.28
0.90	1.812	1.725	1.697	1.645
0.95	2.228	2.086	2.042	1.96
0.99	3.169	2.845	2.750	2.58

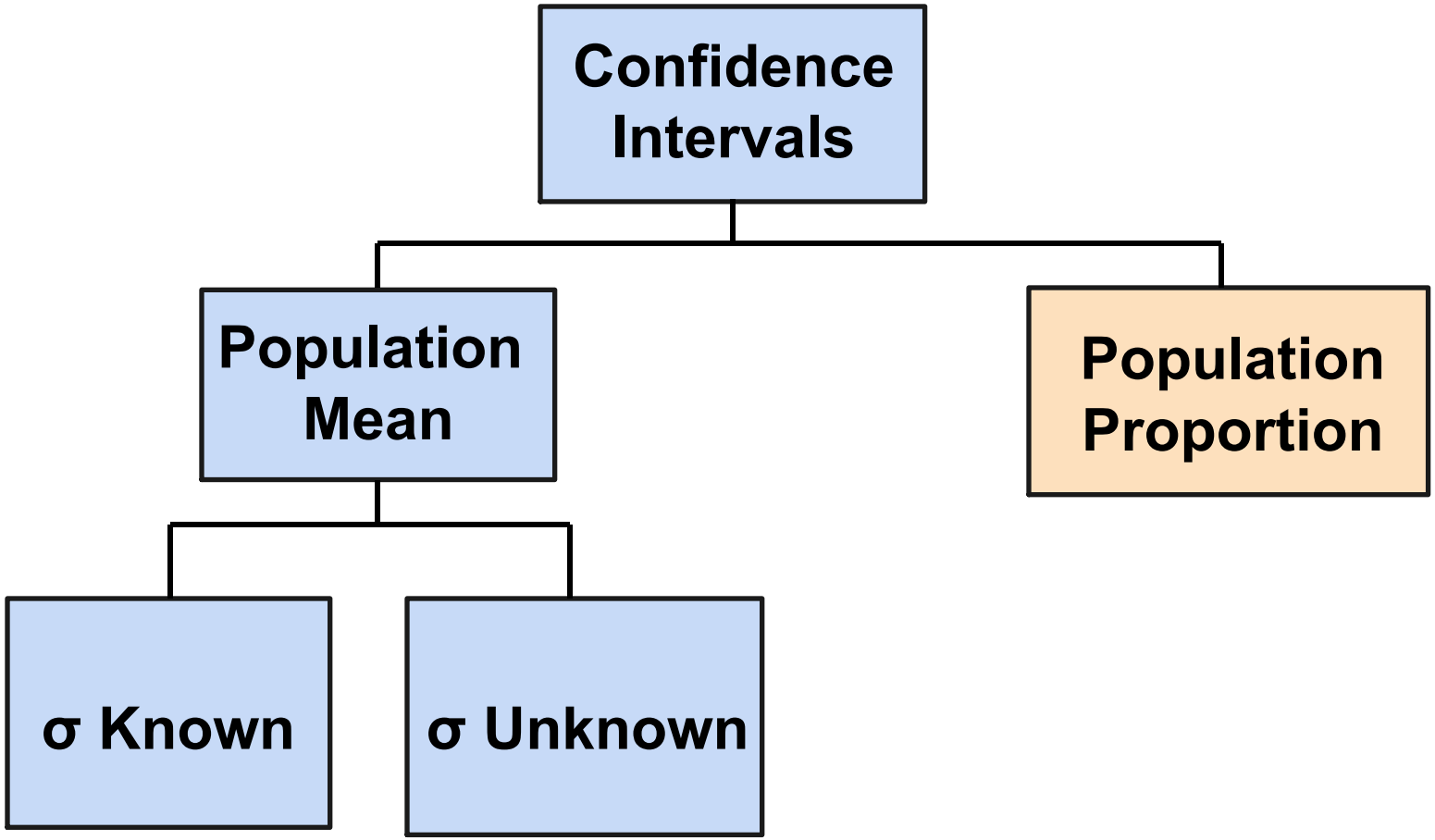
Note: $t \longrightarrow Z$ as n increases

Example

A random sample of $n = 25$ has $\overline{X} = 50$ and $S = 8$. Form a 95% confidence interval for μ

Solution:

Confidence Intervals



Confidence Intervals for the Population Proportion, π

- An interval estimate for the population proportion (π) can be calculated by adding an allowance for uncertainty to the sample proportion (p)

Confidence Intervals for the Population Proportion π

(continued)

- Recall that the distribution of the sample proportion is approximately normal if the sample size is large, with standard deviation

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

- We will estimate this with sample data:

$$\sqrt{\frac{p(1-p)}{n}}$$

Confidence Interval Endpoints

- Upper and lower confidence limits for the population proportion are calculated with the formula

$$p \pm Z\sqrt{\frac{p(1-p)}{n}}$$

- where
  - Z is the standard normal value for the level of confidence desired
  - *p* is the sample proportion
  - *n* is the sample size

Example

- A random sample of 100 people shows that 25 are left-handed.
- Form a 95% confidence interval for the true proportion of left-handers

Example

Solution:

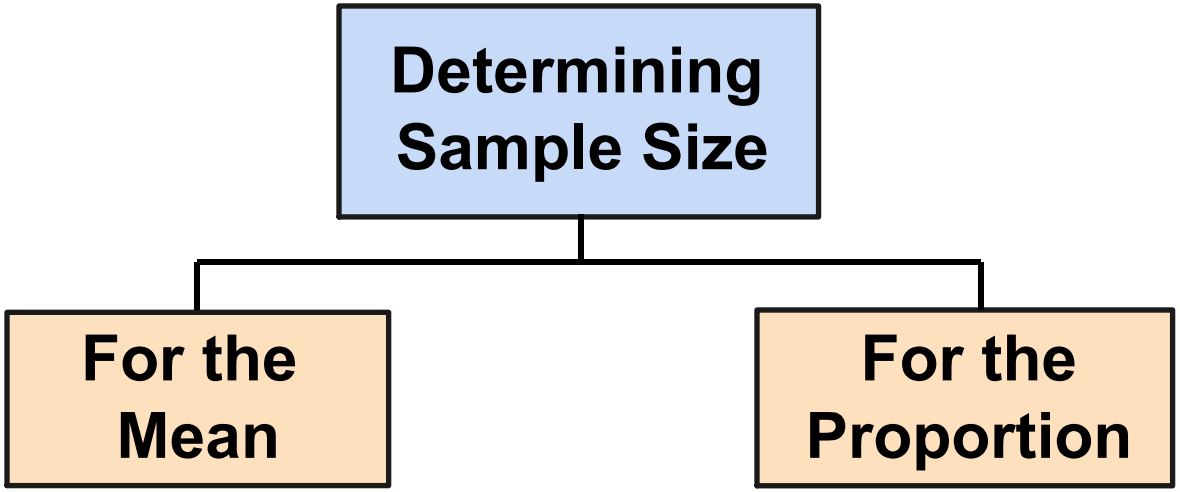
Interpretation

- We are 95% confident that the true percentage of left-handers in the population is between

  16.51% and 33.49%.

- Although the interval from 0.1651 to 0.3349 may or may not contain the true proportion, 95% of intervals formed from samples of size 100 in this manner will contain the true proportion.

Determining Sample Size



Sampling Error

- The required sample size can be found to reach a desired margin of error (e) with a specified level of confidence (1 - α)

- The margin of error is also called sampling error
  - the amount of imprecision in the estimate of the population parameter
  - the amount added and subtracted to the point estimate to form the confidence interval

Determining Sample Size

Determining Sample Size

For the Mean

$$\bar{X} \pm Z\frac{\sigma}{\sqrt{n}}$$

Sampling error (margin of error)

$$e = Z\frac{\sigma}{\sqrt{n}}$$

Determining Sample Size

(continued)

Determining Sample Size

For the Mean

$$e = Z\frac{\sigma}{\sqrt{n}}$$

Now solve for n to get

$$n = \frac{Z^2\,\sigma^2}{e^2}$$

Determining Sample Size

(continued)

- To determine the required sample size for the mean, you must know:
  - The desired level of confidence $(1 - \alpha)$, which determines the critical Z value
  - The acceptable sampling error, e
  - The standard deviation, σ

Required Sample Size Example

If σ = 45, what sample size is needed to estimate the mean within ± 5 with 90% confidence?

Solution:

If σ is unknown

- If unknown, σ can be estimated when using the required sample size formula
  - Use a value for σ that is expected to be at least as large as the true σ
  - Select a pilot sample and estimate σ with the sample standard deviation, S

Determining Sample Size

(continued)

Determining Sample Size

For the Proportion

$$e = Z\sqrt{\frac{\pi(1-\pi)}{n}}$$

Now solve for n to get

$$n = \frac{Z^2\pi(1-\pi)}{e^2}$$

Determining Sample Size

(continued)

- To determine the required sample size for the proportion, you must know:
  - The desired level of confidence $(1 - \alpha)$, which determines the critical Z value
  - The acceptable sampling error, e
  - The true proportion of "successes", π
    - π can be estimated with a pilot sample, if necessary (or conservatively use $\pi = 0.5$)

Required Sample Size Example

How large a sample would be necessary to estimate the true proportion defective in a large population within ±3%, with 95% confidence?

(Assume a pilot sample yields $p = 0.12$)

Required Sample Size Example

(continued)

Solution:

Ethical Issues

- A confidence interval estimate (reflecting sampling error) should always be included when reporting a point estimate
- The level of confidence should always be reported
- The sample size should be reported
- An interpretation of the confidence interval estimate should also be provided

Chapter Summary

- Introduced the concept of confidence intervals
- Discussed point estimates
- Developed confidence interval estimates
- Created confidence interval estimates for the mean (σ known)
- Determined confidence interval estimates for the mean (σ unknown)

Chapter Summary

(continued)

- Created confidence interval estimates for the proportion
- Determined required sample size for mean and proportion settings
- Addressed confidence interval estimation and ethical issues